National Aeronautics and
Space Administration

Synthetic transcriptomics for mouse spaceflight

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BPS

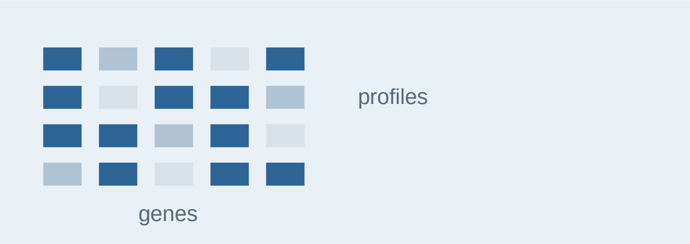
Biological & Physical Sciences

BACKGROUND

Synthetic transcriptomics creates model-generated expression profiles

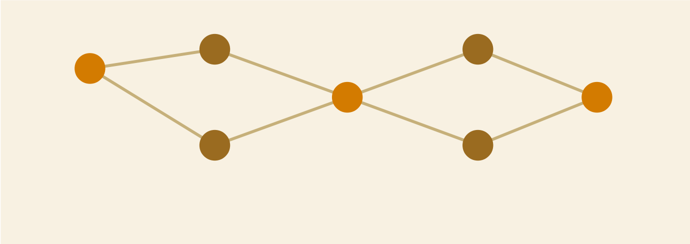
A generator learns patterns in measured RNA-seq data, then samples new expression vectors under chosen conditions.

01 Observed profiles



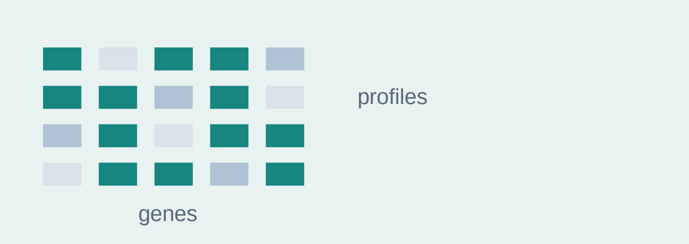
Measured gene expression from real mouse tissue

02 Learn the distribution



Capture gene relationships and variation by tissue and FLT/GC condition

03 Synthetic profiles



Sample new numeric expression vectors for a selected context

It can help test classifiers, rank candidate genes and compare matched FLT and GC profiles

It does not add new animals, new measurements or independent biological evidence

QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

Observed spaceflight cohort

1,610 profiles

75 accessions

835 FLT

775 GC

ARCHS4 mouse

Reference pretraining cohort

997,515 profiles screened → **17,244** selected

20 tissues

Study effects can look like spaceflight biology.

01 Preserve tissue and FLT/GC structure.

02 Avoid memorizing individual profiles.

03 Test biological effects in observed OSDR samples.

Sources: NASA OSDR Biological Data API; ARCHS4 mouse v2.5.

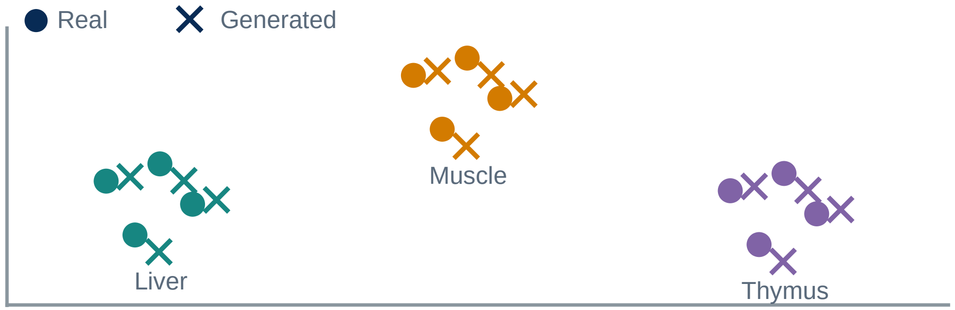
STUDY GOALS

Match tissue distributions, then test FLT versus GC biology

The first goal validates the generator. The second asks whether synthetic data improves a real-data analysis.

01 Reproduce tissue structure

Synthetic bulk RNA-seq should occupy the same tissue-defined expression space as real profiles.



02 Preserve the FLT/GC difference

Within each tissue, generated profiles should retain the smaller condition signal.

OUTPUT

Tissue-specific genes and pathways

Model choice Use the pipeline that matches tissue structure and performs best on held-out real FLT/GC samples.

Gene effects and BH FDR use observed OSDR data.

METHOD

We built a configurable bulk RNA-seq generation pipeline

The downstream analysis used the outlined options; the remaining choices stay configurable.

01 Data scope

Sources

- **OSDR API**
- **ARCHS4**

Studies

- Single
- **Multiple**

Tissues

- Per tissue
- **All tissues**

02 Transform

Expression

- Raw
- CPM
- **TPM**

Scaling

- None
- Z-score
- **MaxAbs**

Features

- All
- HVG
- **L1000**

03 Harmonization

Global or study correction

- **None**
- Within-study z-score
- ComBat / MBatch
- MOBER

04 Model training

Generator

- WGAN-GP
- **DDIM**

Training source

- OSDR only
- ARCHS4 only
- **Pretrain + adapt**

05 Conditioning

Model inputs

- **Tissue**
- **FLT / GC**
- **Accession**
- **Material type**
- Sex / age

Downstream configuration

PREPROCESS: TPM / MaxAbs / 974 landmarks → PRETRAIN: ARCHS4 → ADAPT: OSDR → GENERATE: Conditional DDIM

CONDITION: **Tissue** **FLT / GC** **Accession** **Material type**

HARMONIZE: **None**

SCOPE: **All tissues**

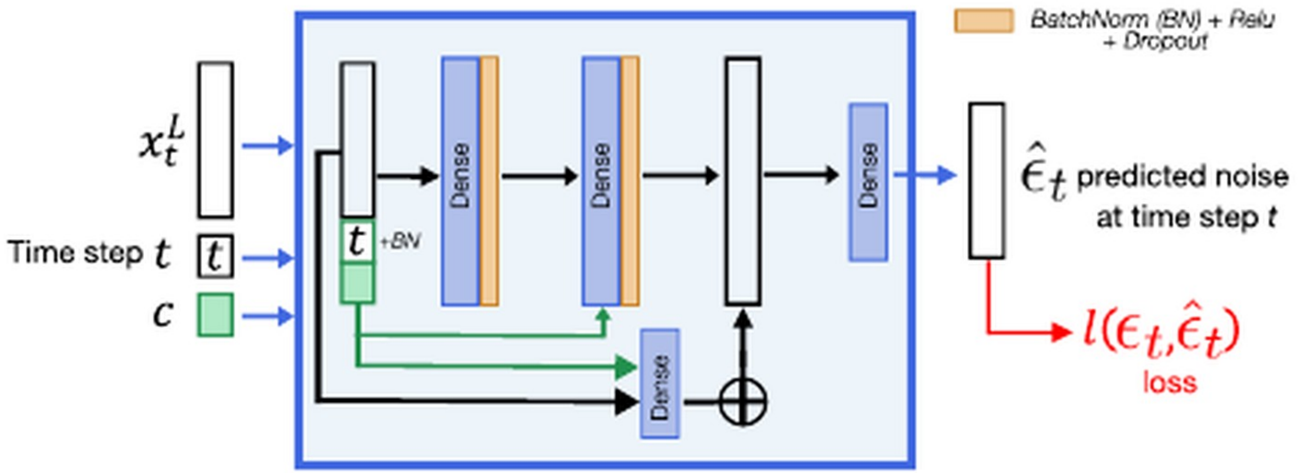
Screen: 463 planned configurations and nine matched liver harmonization arms. Models: Vinas et al. (2022); Lacan et al. (2026).

VALIDATION

DDIM had lower separability and distributional distance

Both generators matched expression. DDIM was harder to distinguish from real profiles and had higher F1.

Architecture from Lacan et al.



Residual dense denoiser conditioned on timestep and tissue.

OSDR adaptation adds FLT/GC, accession and material context through LoRA.

ARCHS4 tissue probe Real 0.781 | Synthetic 0.781

Balanced accuracy is preserved in generated tissue labels.

Metric	WGAN-GP	DDIM	Read
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial acc.	0.636	0.475	near 0.5
FD / real P95	0.144	0.074	lower

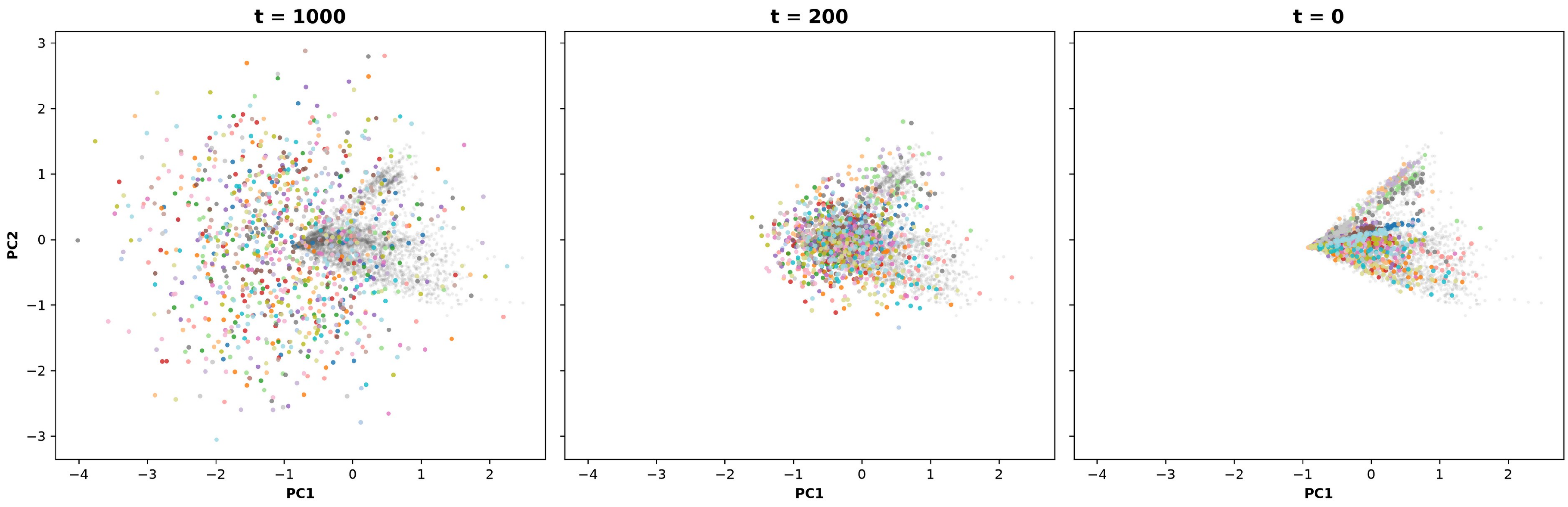
Use DDIM Higher F1 with lower separability and FD.

AA near 0.5 means real and generated profiles are difficult to separate.

Architecture excerpt: Lacan et al. (2026), Fig. 1C, doi:10.1186/s12859-026-06470-8. Metrics use each model's stated split and are not paired.

Diffusion learns tissue structure from noise

The same PCA axes follow 1,024 generated profiles through reverse diffusion.



t = 1000: noise

t = 200: partial structure

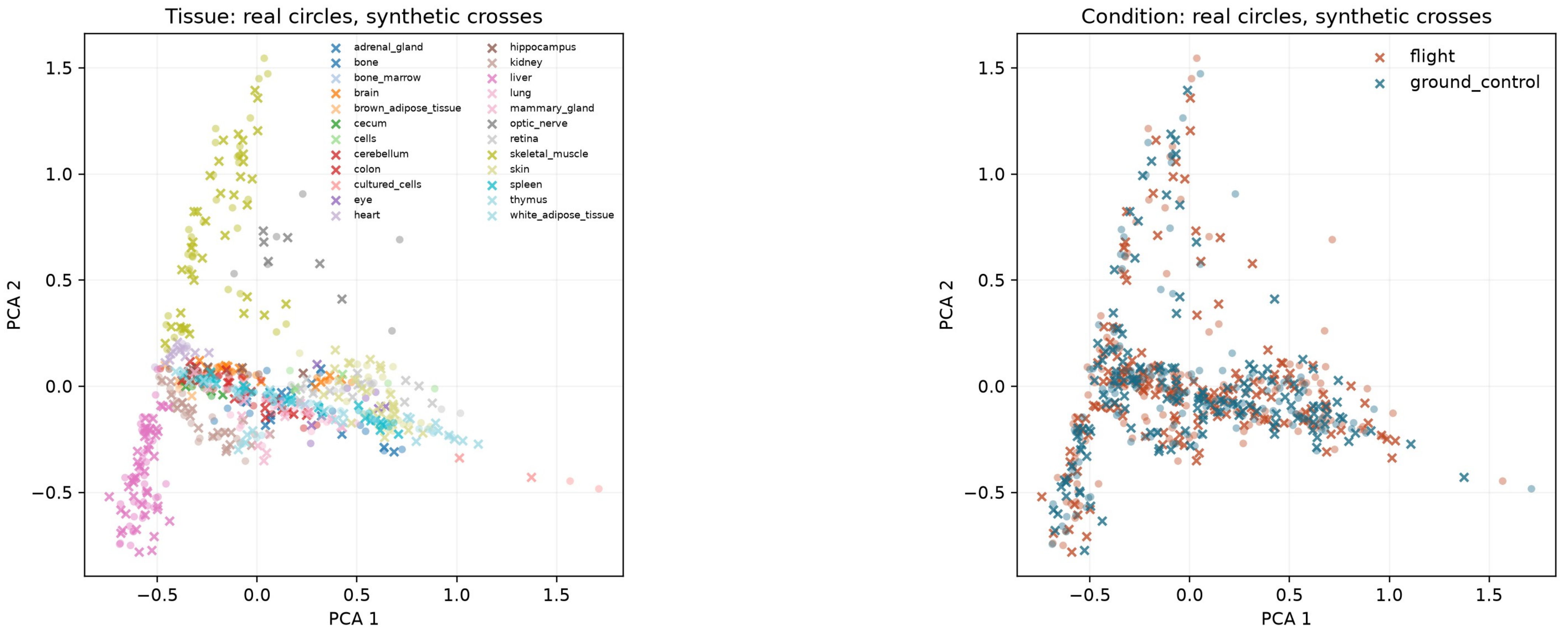
t = 0: tissue-conditioned profiles

Gray points are real ARCHS4 profiles; colors identify generated tissue conditions. PC1 and PC2 limits are shared across panels.

DIFFUSION OUTPUT

Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.



Tissues separate clearly here. Flight and ground-control samples overlap much more, so we test that difference statistically.

PCA was fitted in the common locked OSDR expression space. This is PCA, not UMAP.

ANALYSIS

Five arms separate gene ranking from classifier fitting

Guided arms use synthetic profiles to help choose genes; they do not treat generated profiles as new animals.

INPUTS R real OSDR S matched DDIM GUIDED R + S help rank genes 5% S carries 5% of total fit weight

ANALYSIS ARM	GENE RANKING	CLASSIFIER FIT	WHAT IT TESTS
Real only	R → Rank from real	R → Fit real	Baseline
Generated only	S → Rank from generated	S → Fit generated	Synthetic-only stress test
Real + generated	R + S → Consensus rank	R + S → Equal total weight	Direct augmentation
Guided: real fit	R + S → Consensus rank	R → Fit real only	Feature guidance only
Guided: 5% synthetic	R + S → Consensus rank	R + S → S = 5% weight	Guidance + light regularization

All five arms → Held-out real profiles → BA | AUROC | AP → Choose an eligible arm within each tissue

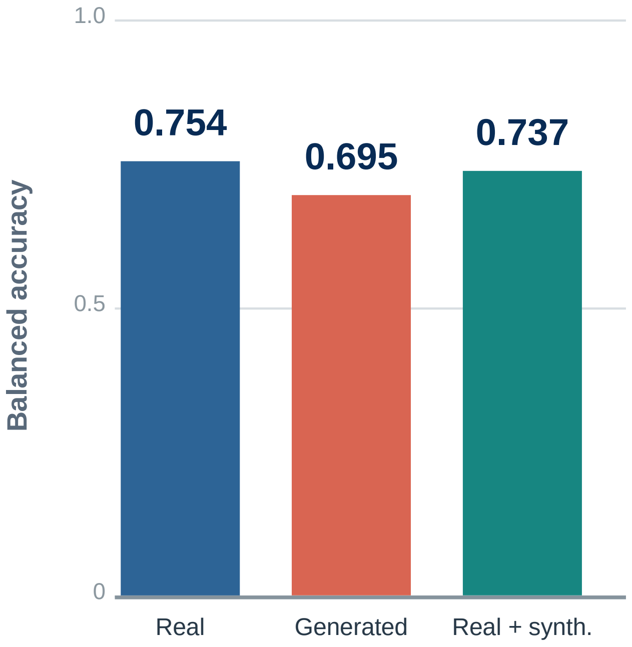
Association test FLT vs GC effects and BH FDR use real OSDR profiles only. **Animal n stays unchanged.**

UTILITY

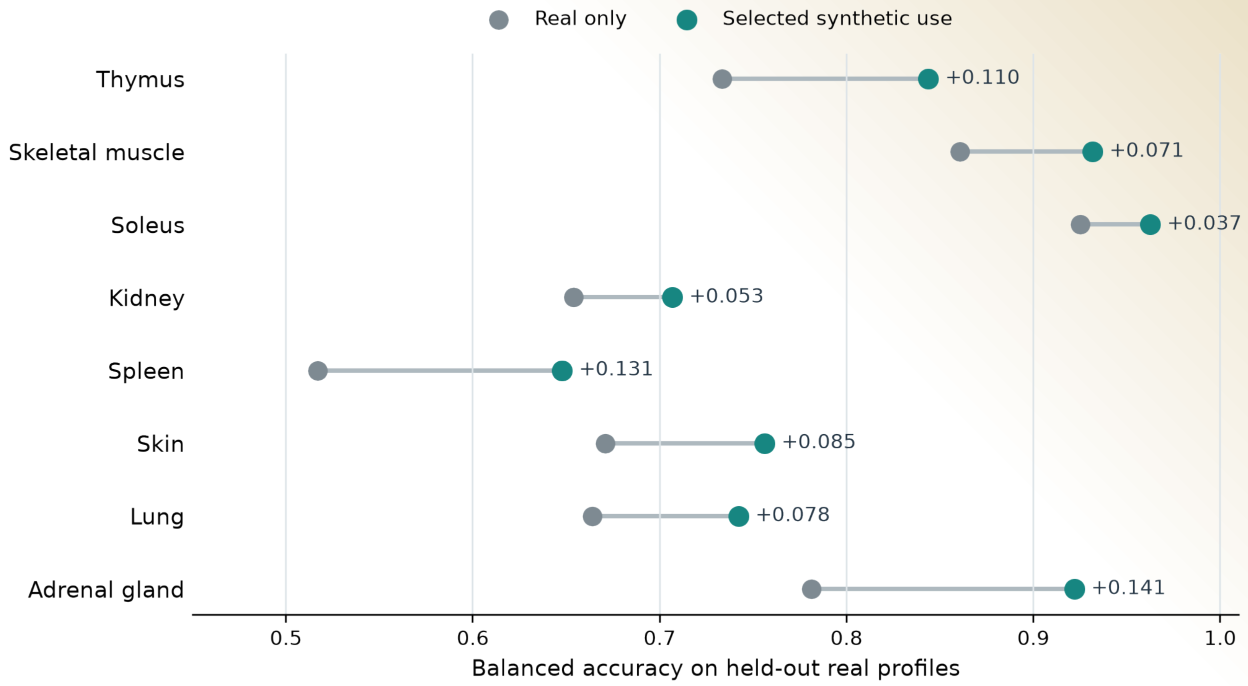
Pooled training missed improvements seen within tissues

The pooled FLT/GC classifier declined, while several tissue-specific classifiers improved.

Pooled across tissues



Selected use within each tissue



The useful arm differed by tissue.

Development results with held-out profiles from represented accessions; they do not measure transfer to a new mission.

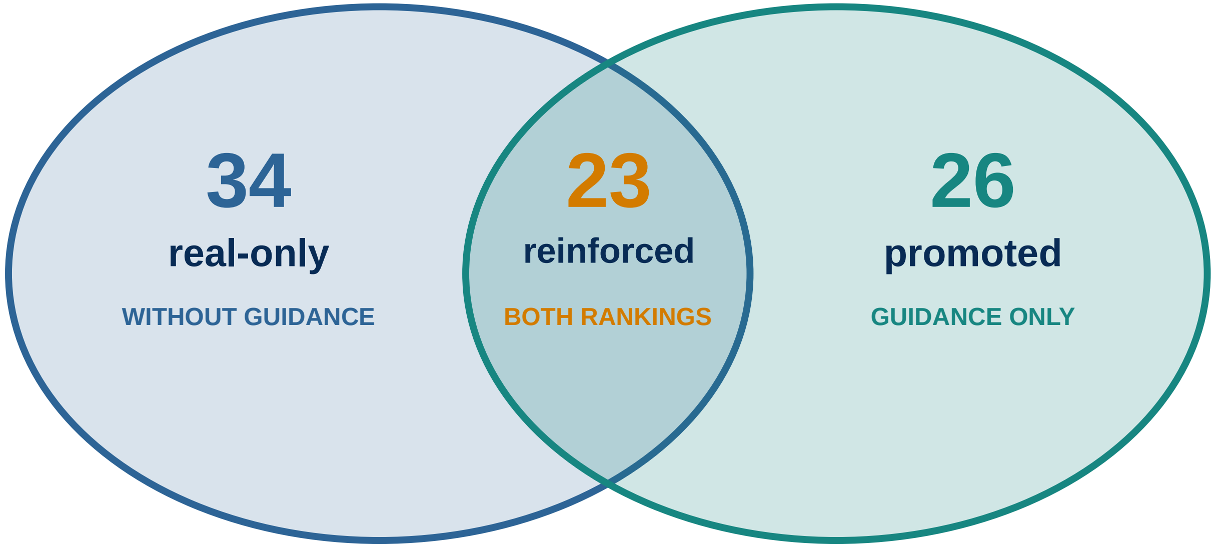
FEATURE ANALYSIS

Synthetic guidance changed ranking, not statistical evidence

Blue marks real-only selection; teal marks synthetic-guided selection. Every displayed association has real OSDR support.

Stable selection after real-data support

- Blue: stable in real-only ranking
- Teal: stable in synthetic-guided ranking



Real OSDR evidence gate

- Estimate FLT vs GC inside each accession
- Combine accession effects with a random-effects model
- Apply BH FDR within each tissue

23 reinforced + 26 promoted

49 synthetic-informed associations

The 34 real-only associations remain real-data findings; synthetic guidance did not reinforce them.

Counts are tissue-gene associations after BH FDR and compatible real-effect direction; genes may recur across tissues.

LITERATURE INTERPRETATION

Selection and literature are separate dimensions

All 49 synthetic-informed associations received both a selection status and a literature interpretation.

01 Fix the candidate
Gene, tissue, and FLT direction

02 Search
Spaceflight and mechanism literature

03 Compare
Observed result with published evidence

04 Record
Category, rationale, and source IDs

Selection status

26 Promoted
Stable only with synthetic guidance

23 Reinforced
Stable with and without guidance

Literature interpretation

Class	Count	Meaning
Aligning	22	Direct or same-tissue process agreement
Complementary	19	Related process or mechanism
Ambiguous	4	Mixed or context-dependent evidence
Unmatched	4	No sufficiently specific match found

A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.

Table S16 records gene-level rationale, evidence scope and source IDs; Table S17 provides 33 citations with DOI/URL and data relationship.

ANALYSIS COVERAGE

The screen covered all 27 completed tissue analyses

The biological discussion narrows only after reporting synthetic-informed, real-only, and null outcomes.

10 Synthetic-informed

Promoted or reinforced BH-FDR gene

ANALYSIS UNITS

- Adrenal Gland
- Eye
- Kidney
- Skeletal muscle (pooled)
- Skin
- Spleen
- Thymus
- Gastrocnemius
- Soleus
- Tibialis Anterior

5 Real BH-FDR only

Real association, no synthetic-informed selection

ANALYSIS UNITS

- Heart
- Liver
- Retina
- EDL
- Quadriceps

12 No BH-FDR gene

No BH-FDR gene in the 974-gene panel

ANALYSIS UNITS

- Bone
- Bone Marrow
- Brain
- Brown Adipose Tissue
- Cecum
- Cerebellum
- Colon
- Hippocampus
- Lung
- Mammary Gland
- Optic Nerve
- White Adipose Tissue

Twenty-two canonical tissues plus EDL, gastrocnemius, quadriceps, soleus and tibialis anterior; complete counts in Table S12.

GENE INVENTORY

Ten tissue analyses contained synthetic-informed genes

All 49 associations passed BH FDR in real OSDR data; synthetic profiles affected prioritization, not the statistical test.

Rows: FLT direction + status — aligning — complementary — ambiguous — unmatched

Gene color = literature | Non-empty rows shown

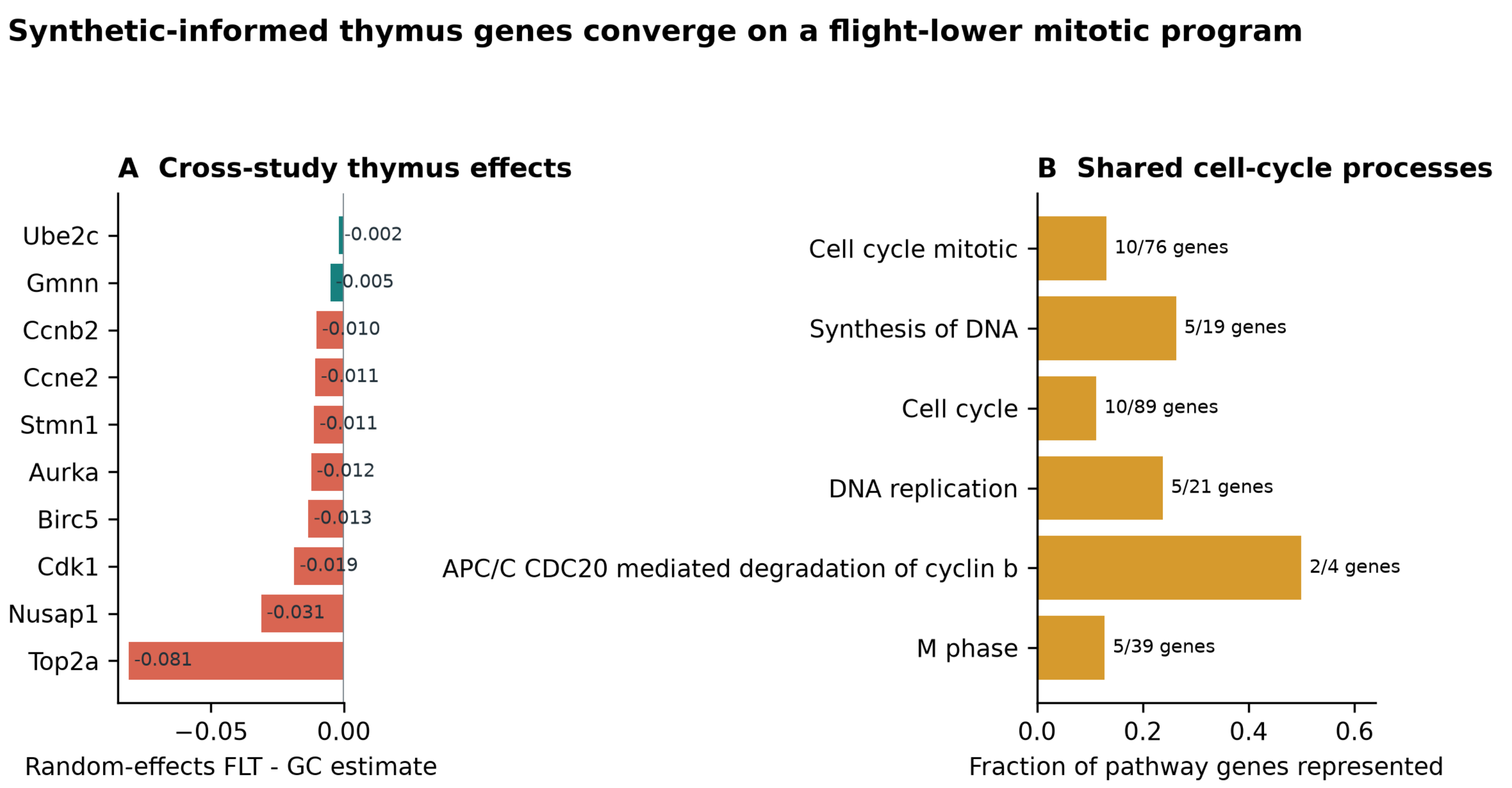
Tissue	FLT direction	Status	Genes
Thymus 16 genes	FLT higher	Promoted	*Hsd17b11*, *Etv1*
	FLT higher	Reinforced	*Snx7*
	FLT lower	Promoted	*Nusap1*, *Stmn1*, *Birc5*, *Cdk1*, *Top2a*, *Ccnb2*, *Aurka*, *Ccne2*, *Kif20a*, *Pcna*, *Ccnf*
	FLT lower	Reinforced	*Ube2c*, *Gmnn*
Spleen 4 genes	FLT higher	Promoted	*Rai14*, *Myl9*, *Ptprk*
	FLT higher	Reinforced	*Loxl1*
Kidney 2 genes	FLT higher	Promoted	*Inpp4b*
	FLT higher	Reinforced	*Slc37a4*
Gastrocnemius 2 genes	FLT higher	Promoted	*Nfkbia*
	FLT lower	Promoted	*Fhl2*
Eye 1 gene	FLT lower	Reinforced	*Klhl21*
Skeletal muscle (pooled) 12 genes	FLT higher	Reinforced	*Sox4*, *Cebpd*, *Sh3bp5*, *Prkcd*, *Arid5b*, *Sesn1*, *Tle1*
	FLT lower	Promoted	*Klhl21*, *Mapkapk5*, *Reep5*, *Itgb5*
	FLT lower	Reinforced	*Bphl*
Soleus 5 genes	FLT higher	Reinforced	*Tpm1*
	FLT lower	Reinforced	*Bdh1*, *Ech1*, *Bnip3*, *Decr1*
Tibialis anterior 4 genes	FLT higher	Promoted	*Cebpd*
	FLT higher	Reinforced	*Cdkn1a*, *St3gal5*, *Bnip3*
Adrenal gland 2 genes	FLT lower	Promoted	*Psmb8*
	FLT lower	Reinforced	*Tspan4*
Skin 1 gene	FLT higher	Promoted	*Plscr1*

Separate rows report FLT direction and selection status; gene color independently reports literature interpretation for all 49 associations.

THYMUS

Thymus points to lower proliferative renewal

The cell-cycle program persisted without material conditioning; two FLT-higher genes extend the hypothesis.



16 synthetic-informed BH-FDR associations

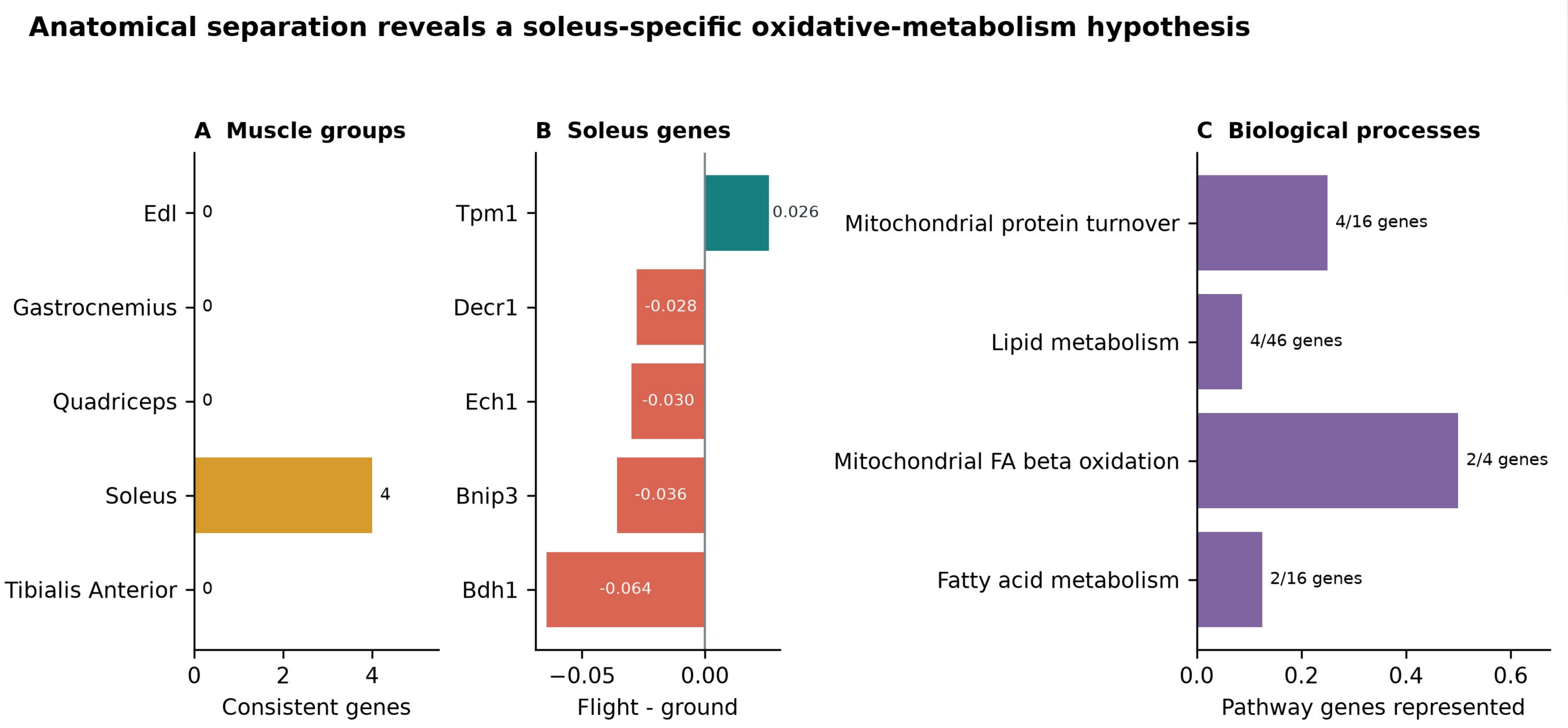
13 promoted | 3 reinforced

- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

Context: Gridley et al. (2013), Horie et al. (2019), Keenan et al. (2025), and Shi et al. (2026).

Soleus reinforces a mitochondrial and lipid program

The real-data program is coherent, but its synthetic reinforcement required explicit material conditioning.



0.925 to 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

Literature context: Gambara et al. (2017) and Stein et al. (2002).

Additional tissues produced distinct hypotheses

Promoted and reinforced selections are shown separately for each tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Pooled muscle	Promoted	Lower: Klhl21, Mapkapk5, Reep5, Itgb5	Heterogeneous stress, differentiation, interferon and sialic-acid response; interpret with the anatomical muscle groups.
	Reinforced	Higher: Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1; lower: Bphl	
Kidney	Promoted	Inpp4b higher	Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.
	Reinforced	Slc37a4 higher	
Spleen	Promoted	Rai14, Myl9, Ptprk higher	Adhesion, actomyosin and extracellular-matrix or immune-organization hypothesis; no coherent pathway enrichment.
	Reinforced	Loxl1 higher	
Skin	Promoted	Plscr1 higher	Interferon-linked skin candidate within broader cell-cycle and DNA-repair responses; a single-gene result.
	Reinforced	None	

All listed effects passed BH FDR in observed OSDR profiles; 'None' means no gene in that selection category.

ADDITIONAL FINDINGS II

Eye, adrenal and muscle-group results remain tissue-specific

Promoted and reinforced selections are separated within every tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Eye	Promoted	None	Process-level alignment with lower proliferation and cytokinesis in flight eye tissue; not an exact prior gene replication.
	Reinforced	Klhl21 lower	
Adrenal gland	Promoted	Psmb8 lower	Literature-unmatched immune, proteostasis or tissue-composition candidates; no adrenal mechanism is established.
	Reinforced	Tspan4 lower	
Gastrocnemius	Promoted	Nfkbia higher; Fhl2 lower	NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.
	Reinforced	None	
Tibialis anterior	Promoted	Cebpd higher	Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.
	Reinforced	Cdkn1a, St3gal5, Bnip3 higher	

Interpretations follow the independent literature annotations in Table S16; 'None' means no gene in that selection category.

TAKEAWAYS

Synthetic data worked best as a tissue-specific prior

The generator helped prioritize observed signal. Biological claims still come from OSDR profiles.

01 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

02 Use

Pooled augmentation failed. Tissue-specific ranking and low-weight training were more useful.

03 Interpret

Thymus was robust to material ablation; soleus was conditioning-sensitive. Literature review separated recovery from extension.

Next test

Use independent samples and cell-resolved assays to test thymus proliferation, soleus metabolism and prioritized candidates from additional tissues.

Generated profiles never enter the biological sample count or the BH-FDR test.

Thank you

Questions?

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Data	NASA OSDR | ARCHS4 | Reactome
Compute	NASA Ames Research Center
Code and manuscript	github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.